

00960222: Language, Computation and Cognition

Project Guidelines

Spring 2026

Students enrolled in 00960222 are required to complete a class project which constitutes 80% of the course grade. The project is due **July 14** at 23:55. You will work on the project in pairs. In very exceptional cases and upon TA approval, you may work on the project in a team of three (with the expectation of an extended project scope). You are expected to work on the project continuously throughout the semester. As a rule of thumb, a good project requires at least 50 hours of work from each project member, so you should plan on dedicating several hours per week for the project during the rest of the semester. The project deliverables are (1) a project report, (2) a link to a Github repository with your code, and (3) an in-class project presentation during the last week of the semester.

Project Tracks (Eyes and Brain) By default, you are expected to carry out either Eyes Track or Brain Track as proposed below. Each track is built around a community benchmark.

Open Projects (for PhD students) PhD students whose research is in NLP, psycholinguistics or related areas can (but are not obligated to) do an open research project on a topic of their choosing. The project should be *clearly* related to the topics covered in the course, and will need to be approved by the course staff. If you would like to do such a project, you are asked to submit a 1-page project proposal by **June 4** and meet with the course staff.

Project Reports All projects should include a written report, which should follow from the content of the project. The report length should be **4–6 pages**, with an upper limit of 8 pages, comparable to an ACL short/long paper. Please use the Cognitive Science¹ or ACL² conference templates for your writeup.

The report is an integral component of the project, and *part of the grade will be given based on the quality and the clarity of the report*. You should think of the report as a scientific paper that you are submitting to a scientific conference like ACL or CogSci, to be evaluated via peer review. You can expect the reviewer to be generally literate in computational psycholinguistics, but you cannot rely on them having any specific knowledge of your work.

Report requirements and recommendations:

¹<https://cognitivesciencesociety.org/submissions/>

²<https://github.com/acl-org/acl-style-files?tab=readme-ov-file>

- Report sections:
 - Abstract: a short, one paragraph summary of the main scientific and/or engineering questions that the work addresses and the main conclusions. At the end of the abstract you should have a *footnote with a link to a GitHub repository with your code*.
 - Introduction: A more elaborate exposition of the scientific background, key relevant prior work with citations, your research question(s) and why it is novel and/or interesting and your main findings.
 - Data: A short description of the dataset(s) you worked with, with key statistics such as the number of words in a textual corpus, number of participants in a human subjects experiment, etc.
 - Experiments and Results: This is the heart of the report which should describe in detail (i) *what you did*, (ii) *how you did it*, and (iii) *the results of your experiments / analyses and what we learned from them*. This section may be divided into multiple parts as you see fit, and should constitute most of the writeup. If you are using an approach that relies on previously published papers or are using code from published papers, be sure to cite them. Figures and tables are generally helpful to include in this part of the report. Figure axes should be clearly labeled and a legend should be included when relevant. Each figure and each table should include an informative caption that explains it and briefly states the conclusion(s) that can be drawn from it. Every figure and table must be explicitly referenced and discussed in the main text. Information that appears in captions can be reiterated in the text.
 - Discussion and Conclusions: This section should summarize the key results and discuss their theoretical and/or practical implications, as well as the limitations of the work. Directions for future work may also be included in this section.
 - Bibliography
- Failure to include the sections above and their respective content will lead to deduction of points.
- It is important that your report is well structured and *clearly written*. While we will not deduct points for English grammar mistakes, if the exposition of the work is sloppy, unclear or partial, this will lead to deduction of points.
- You may include an appendix with additional figures and tables if needed. Appendix tables and figures should have captions and should be mentioned and referenced in the main text. The appendix can only contain supplementary materials which are not strictly required for understanding your work. Any results that are key to understanding your work and its contributions should appear in the main text.

- While we ask that you link to the project’s GitHub repository, you cannot expect the reader to consult your raw code in understanding your writeup.
- Before writing the report, we recommend going through a few short/long papers from the proceedings of ACL or CogSci in order to see the exposition style and content type expected in your report.
- We expect that you *proofread* the report before submission.
- We highly recommend starting to work on the report early, at least a month before the submission deadline.

Class Presentation The last week of the semester is dedicated to in-class project presentations. Each team will give a short talk on their project, and both team members are expected to speak. The presentation is part of the project grade. Students should bring their reports to the talk; questions may address the report and are not restricted to the contents of the talk.

Presentation guidelines:

- **Length:** 7 minutes of presentation followed by 7 minutes of questions. Exact timing will be confirmed closer to the date based on the number of teams.
- **Suggested structure:**
 - Motivation and research question (1–2 slides).
 - Data and task (1–2 slides).
 - Method (1–2 slides).
 - Results (2–3 slides).
 - Key takeaways (1–2 slides).
 - Limitations and future work (1 slide).
- **Slides:** prepare slides and upload them (or a link to them) to Moodle.
- **Audience:** assume the audience is your classmates and the course staff, that is, people who are broadly familiar with computational psycholinguistics but not with the specifics of your project.

Grading The expectations from all the projects are *high*, with the following rule-of-thumb division to grade categories:

96-100	Outstanding project with in-depth work that can form the basis for an international conference paper (e.g. ACL / SCiL / CogSci).
90-95	Excellent project with solid work that can form the basis of an international workshop paper (e.g. HSP / CMCL).
80-89	A very good project with non-trivial analyses and adequate scope.
70-79	A good project with notable quality issues and/or limited scope.
60-69	A project with major quality issues and/or very limited scope.
55-59	A project that meets the minimum requirements.

Note that the project type (Eyes Track / Brain Track / Open) has *no impact on the grade*. Further note that negative experimental results are just as good as positive results; grading will be based only on the quality of the ideas, the level of execution in the analyses, the scope of the project and the quality of the report.

Timeline

- **by June 4 (23:55)** team up with a partner. Register your team on Moodle and indicate the track (Eyes or Brain) you are planning to work on.
 - Project Tracks (Eyes or Brain): No further action is required.
 - Open Projects (PhD students only): Send us (Yevgeni, Omer and Keren) an email attaching a 1-page project proposal. In your email, include your availability for a 20 minutes meeting to discuss project details.
- **by June 21 (23:55)** all projects: submit on Moodle a 1 page progress report. The report should specify your progress so far and a concrete plan for the remaining work, including your research question and planned analyses. Projects in good standing should ideally have preliminary results by this point.
- **Week of June 23** The recitation time will be dedicated to project office hours with the course staff.
- **by July 14 (23:55)** Submit the final report on Moodle.
- **Week of July 21** In-class project presentations.

Late Submissions Policy

Teams that do not submit the June 21 progress report will be deducted 4 points from the project final grade. Late submissions of the progress report will be deducted 1 project final grade point per day (up to 4 days). Final project reports submitted after the July 14 deadline will be deducted 3 points per day, up to a maximum of 5 days. Projects submitted over five days after the deadline will not be graded. Projects that are not presented in class will receive a fail grade for the project, and consequently a fail grade for the course.

Project Tracks

All projects this year are built around community benchmarks in computational psycholinguistics and cognitive science: Track Eyes focuses on predictive modeling from *eye movements in reading*, and Track Brain on AI models of *neural brain signals*. Pick one of the two tracks below, and within it choose at least one dataset, choose at least one task, and propose and address a research question that builds on the data, the task, the model, or their interactions.

Eyes Track

EyeBench: Predictive Modeling from Eye Movements in Reading

Eye movements in reading provide a fine-grained, temporally detailed record of how readers interact with text, and offer valuable insights into the reader, the text, and the cognitive processes that underlie language comprehension. EyeBench (Shubi et al., 2025) is a benchmark for decoding cognitive and linguistic information from eye movements during reading. Version 1.0 includes 7 prediction tasks across 6 harmonised datasets, 12 implemented models, and 3 evaluation regimes. The EyeBench paper and project page are available at <https://eyebench.github.io/>; we expect you to read the paper before designing your project.

Propose and address a research question that builds on one or more of the datasets, tasks, models, or evaluation regimes. The available datasets, tasks, models and evaluation regimes are listed in Sections A.1–A.4 below. We recommend going over all four sections before settling on a direction. Your project must include a substantial analytical component such as an ablation study, error analysis, feature analysis, model comparison, or a modification to the training procedure or model architecture; simply running an existing model is not sufficient. If applicable, use the benchmark’s official splits and metrics for evaluation. You do not have to work on one of the 7 prediction tasks: it is equally acceptable to use the data to address any other cognitive or linguistic question of interest.

A.1 Datasets

EyeBench v1.0 includes 6 datasets: OneStop (Berzak et al., 2025), SB-SAT (Ahn et al., 2020), PoTeC (Jakobi et al., 2025), MECO L2 (Kuperman et al., 2023, 2025), CopCo (Björnsdóttir et al., 2023; Hollenstein et al., 2022; Reich et al., 2024), and IITB-HGC (Maharaj et al., 2023). See the appendix of Shubi et al., 2025 for summary statistics and further details, and refer to the original dataset papers cited above for the full descriptions of each corpus.

A.2 Tasks

- **Reading Comprehension** (OneStop, SB-SAT, PoTeC): binary classification of whether the reader answered a multiple-choice comprehension question correctly.
- **Reading Comprehension Skill** (CopCo): regression on the reader’s overall reading comprehension score (1–10).
- **Vocabulary Knowledge** (MECO L2): regression on the reader’s LexTALE (Lemhöfer & Broersma, 2012) vocabulary test score (0–100).
- **Dyslexia Detection** (CopCo): clinical classification of whether the reader has dyslexia.
- **Subjective Text Difficulty** (SB-SAT): Likert-scale regression on the reader’s self-reported difficulty rating of the read text.

- **Domain Expertise** (PoTeC): classification of whether the reader is a domain expert (physicist vs. biologist) for the passage being read.
- **Claim Verification** (IITB-HGC): classification of whether the reader correctly judged the truth of a claim about the text.

A.3 Models

You may extend or modify one of the EyeBench models listed below, port a model from outside the benchmark, or train a new one. The 12 models implemented in EyeBench, grouped by model family:

- **CNN-based:** AhnCNN.
- **RNN-based:** AhnRNN, BEyeLSTM, PLM-AS, PLM-AS-RM.
- **Transformer-based:** RoBERTEye-W, RoBERTEye-F, MAG-Eye, PostFusion-Eye.
- **Traditional ML model:** classical machine-learning models (e.g., logistic regression, gradient boosting) trained on aggregated eye-movement features.

Table 2 of the EyeBench paper (Shubi et al., 2025) is a useful starting point, as it emphasises the different eye-movement representations available to models.

A.4 Evaluation Regimes

EyeBench evaluates models under 3 regimes:

- **Unseen reader:** the model is tested on readers that were not seen during training (but on texts that have been seen).
- **Unseen text:** the model is tested on texts that were not seen during training (but on readers that have been seen).
- **Unseen reader and text:** both the reader and the text are held out during training.

Brain Track

NeuralBench: NeuroAI Models for Brain Signals

Brain signals such as EEG, MEG, and fMRI record neural activity during cognitive processing, and provide valuable insights into the neural basis of language comprehension, attention, memory, and more. NeuralBench (Meta AI, 2026) is a unified open-source framework for benchmarking AI models of brain activity. NeuralBench-EEG v1.0 includes 36 EEG downstream tasks across 94 datasets and 14 deep-learning architectures (8 task-specific and 6 foundation models). 2 MEG tasks and 1 fMRI task are also included. NeuralBench also includes `NeuralFetch` for data acquisition, `NeuralSet` for preprocessing and dataloaders, and `NeuralTrain` for PyTorch-Lightning training, which can be useful for your project. The NeuralBench paper and documentation are available at <https://facebookresearch.github.io/neuroai/neuralbench/>; we expect you to read the paper before designing your project.

Propose and address a research question that builds on one or more of the datasets, tasks, or models. The available datasets, tasks and models are listed in Sections B.1–B.3 below. We recommend going over all three sections before settling on a direction. Your project must include a substantial analytical component such as an ablation study, error analysis, feature analysis, model comparison, or a modification to the training procedure or model architecture; simply running an existing model is not sufficient. If applicable, use the benchmark’s standardised splits and task-appropriate metric for evaluation. Projects are not restricted to the NeuralBench decoding tasks; you may also use the data to address any other cognitive or linguistic question of interest.

B.1 Datasets

NeuralBench provides datasets across 3 brain-recording modalities: **EEG** (the main release, with 94 datasets), **MEG** (2 tasks), and **fMRI** (1 task).

B.2 Tasks

Focus on language-related tasks such as sentence decoding, word decoding, speech decoding, and N400 semantic-processing classification. Documentation for each task is available on the NeuralBench Tasks page³, including the list of datasets registered under each task.

B.3 Models

You may extend or modify one of the architectures that ship with NeuralBench, port a model from outside the benchmark, or train a new one. Below we list the architectures that are most relevant to the language-related tasks of Section B.2. NeuralBench includes additional models that are less natural fits for language-related tasks.

³<https://facebookresearch.github.io/neuroai/neuralbench/tasks/tasks.html>

- **CNN-based:** eegnet, simpleconv_time_agg.
- **Transformer-based:** eegconformer, LaBraM, LUNA, REVE.

Notes for both tracks

- **Understand the experimental setup and the available data** before designing your analysis. We highly recommend reading the dataset paper, looking at the data, and running exploratory data analysis and statistics.
- **Look beyond the prediction targets.** Most datasets also include additional meta-data such as participant characteristics (e.g., age, language background, education, dyslexia status) and stimulus-level annotations (e.g., text topic, difficulty level). These can be useful as predictors, as control variables, or as the basis for your own research question.
- **Survey related work.** We encourage you to go over papers that cite the dataset(s) and model(s) you choose. Tools such as Connected Papers and Google Scholar’s “Cited by” view are an efficient way to see what has already been studied with these resources and to find directions worth building on or contrasting against.
- **Your project does not have to be a prediction task.** The official benchmark tasks are one valid option, but you are equally welcome to use the data to address any cognitive, linguistic, or methodological question of interest, with or without a predictive-modeling component.

References

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